MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

1/ TYPE 6995

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, CW, tunable frequency range 16,000 to 17,000 MHz, rated power output 20 mW

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Irs	T(body)	Alt
Unit:	V	Vdc	Vdc	mAdc	°C	ft
Maximum:	6.93	350	0 to -500	45	200 (Note 2)	No limit

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Cooling:	Convection
Cathode position:	Unipotential, oxide coated	Base:	Molded; color coded leads
Mounting:	Any	Output connection:	Mates with UG-419/U

TEST CONDITIONS: See note 2

Parameter:	Ef	Ers	Er	T(body)	F
Unit:	V	Vdc	Vdc	°C	MHz
	6.3	300	Er (Mode 4)/Max Po	160 (max)	F2

Frequency	
F	MHz
F1	16,200
F2	16,500
F3	16,760

GENERAL:

First article sample inspection - Required (See note 1)

1/ Formerly VA-94

6995

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	First article sample inspection	1					
1032	Vibration, mechanical	3	10 G; F = 10 to 1,000 Hz; t = 5 minutes (min)	ΔF(p-p)	---	0.5	MHz
1042	Shock, specified pulse	4	Test condition C	ΔF	---	±4	MHz
4027	Temperature coefficient	-	TA = -55° to +105°C	ΔF/ΔT	-0.2	-0.4	MHz/°C
4028	Barometric pressure, reduced	5	t = 10 seconds	ΔF	---	-4	MHz
4223	Mechanical tuning rate	-	F1 to F3	ΔF/ turns	---	350	MHz/ turns
4302	Frequency change (warmup)	6		ΔF	---	4	MHz
---	Heater voltage sensitivity	-	Ef = 6.0 to 6.6 V	ΔF/ΔEf	---	8	MHz/V
	Quality conformance inspection, part 1	7					
4214	Cathode emission	8,9	Ef = 5.7 V	ΔIk/Ik	---	-15	%
1256	Electrode current (resonator)	-		Irs	25	45	μAdc
4250	Power output	-	F1, F2, F3	Po	20	---	mW
4213	Reflector voltage	-	F1 F3	Er Er	-100 ---	--- -200	Vdc Vdc
4229	Total reflector current	8, 9, 10		Ir	---	5	μAdc
4223	Mechanical tuning range		High frequency Low frequency	F F	16,000 ---	--- 17,000	MHz MHz
4280	Electronic tuning range	11	Er (mode 4)/50% max Po; F3	ΔF	55	---	MHz
	Quality conformance inspection, part 2						
4289	Heater current, nonoperating	-		If	1.08	1.32	A
4280	Electronic tuning range	-	Er (mode 4)/50% max Po; F1	ΔF	55	---	MHz
4215	Reflector voltage modulation coefficient	-	F1 and F3	ΔF/ΔV	1.3	3.5	MHz/V
1031	High-frequency vibration	12		Ir	---	10	μAdc
	Quality conformance inspection, part 3						
---	Life test	-	Group D	t	500	---	hrs
---	Life-test end point:						
4250	Power output	-		Po	15	---	mW

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NOTES:

1. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. In addition, first article sample approval shall include satisfactory demonstration of tube-to-system compatibility. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

2. All tests (except vibration tests) shall be conducted with the TUT rigidly connected to a UG-419/U (MIL-F-3922/53-005) flange on RG-91/U (MIL-W-85/1-089) wavegudie equipment and the TUT load VSWR shall be less than 1.1:1. Forced air cooling shall be used for power inputs above 10 watts.

3. The TUT shall be vibrated in each of the three mutually perpendicular axis, one which shall be parallel to the beam axis of the TUT.

4. The TUT shall be shocked 5 times in each of the three mutual perpendicular planes. The frequency shift after shock in any plane shall not exceed the specified limit.

5. The frequency shall be stabilized at a pressure of 70 mmHg. The pressure shall be increased to 760 mmHg and the frequency at 760 mmHg read within the time specified. The resulting frequency change shall not exceed the limit specified.

6. After a cold start in room ambient temperature at time, t = 0, the frequency drift shall be measured from t = 5 minutes to t = 20 minutes.

7. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

8. This test shall be performed at the conclusion of the holding period.

9. The TUT shall not be oscillating during this test.

10. After 2 minutes with all voltages applied, the total reflector current shall not exceed the specified limits.

11. There shall be no frequency discontinuities or mode hysteresis between the half-power points.

12. The TUT shall be vibrated with the cathode end up. The reflector current shall be recorded. Reflector current bursts shall not exceed the specified limits.

13. Revision letters are not used in the revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

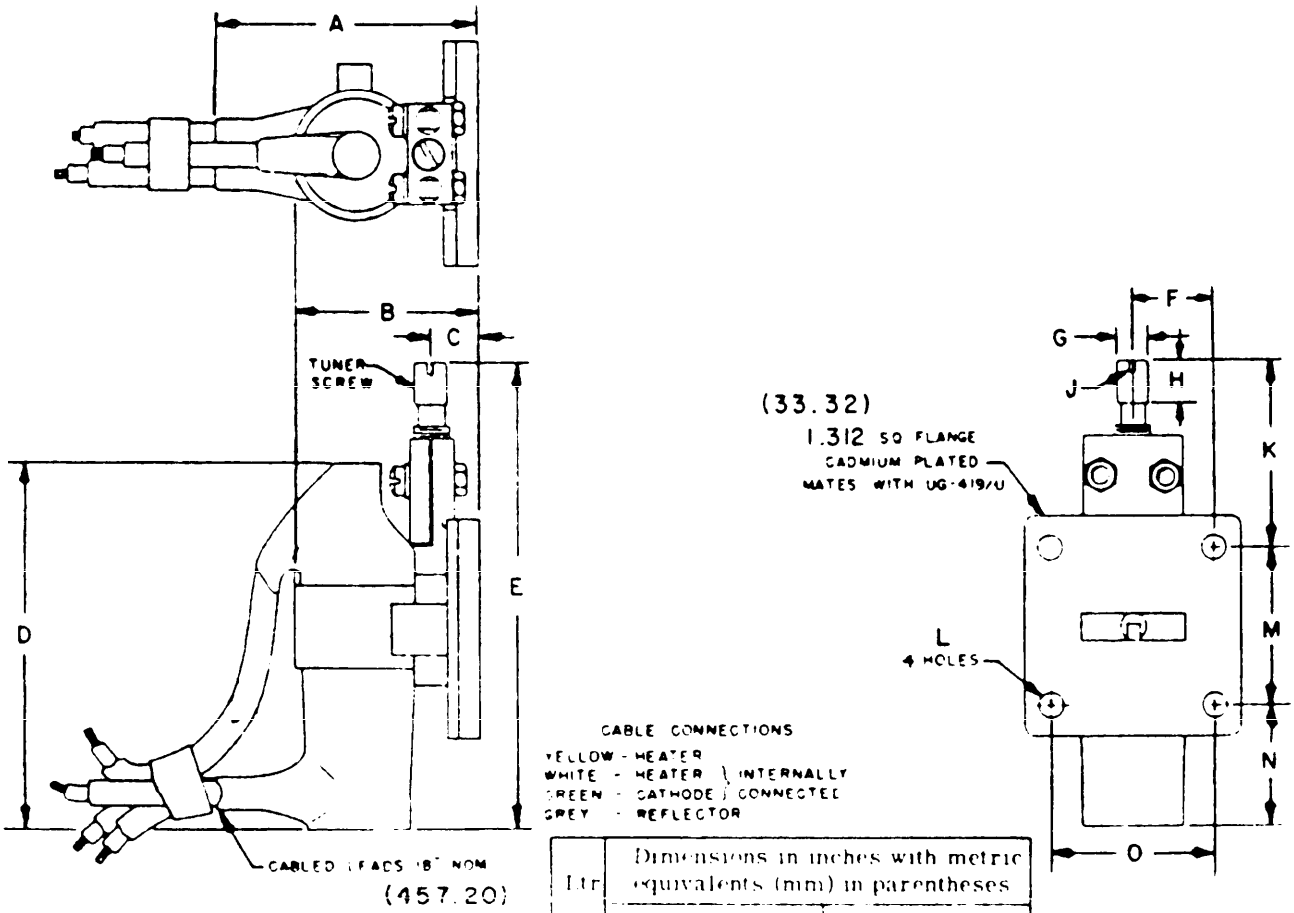
User activities:
Army - ME, WC, MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3066)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note)		
C	.250 (6.35)	.312 (7.92)
E		3.000 (76.20)
G	.185 (4.70)	.187 (4.75)
K	.900 (22.86)	1.200 (30.48)
M	.952 (24.18)	.960 (24.38)
O	.990 (25.15)	.998 (25.35)
Quality conformance inspection, part 2		
A		1.750 (44.45)
D		2.400 (60.96)
F	.466 (11.84)	.528 (13.41)
L	.140 (3.56)	.144 (3.66)
Reference dimensions		
B	1.125 (28.58)	
H	.250 (6.35)	
J	.040 (1.02) wide	.070 (1.78) deep
N	.750 (19.05)	

NOTE: Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level of II.

FIGURE 1. Outline drawing of electron tube type 6995.

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No. 22-R255

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